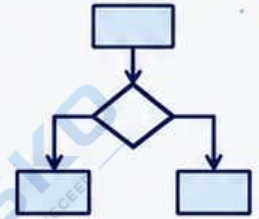


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Most Important Questions

INSTRUCTIONS



1. Understand the Concepts

Go through the basic concepts related to each question before practising.



2. Practice Regularly

Solve these important questions and their variations for better understanding.



3. Focus on Exam Pattern

These questions are selected as per previous year papers and latest syllabus.



4. Attempt Within Time Limit

Practice under timed conditions to improve speed and accuracy.



5. Revise and Self-Check

Review your answers, identify weak areas and revise regularly to score higher.

Learn • Practice • Improve • Succeed

YOUR HARD WORK PAYS OFF!



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Machine Learning — Most Important Questions

Unit I - Introduction & Data Preprocessing

- **Answers in one or two lines (1 mark questions)**

1. What is Machine Learning?
2. Differentiate between Traditional Programming and ML.
3. Name the types of Machine Learning.
4. What is Supervised Learning?
5. What is Unsupervised Learning?
6. What is Reinforcement Learning?
7. What is Data Preprocessing?
8. What is Feature Scaling?
9. What is Train-Test Splitting?
10. What is Pixel Normalization?

- **Short Answer Questions (3 marks questions)**

1. 🔥 Define Machine Learning and explain its applications.
2. 🔥 Differentiate between Traditional Programming and Machine Learning.
3. 🔥 Explain Supervised, Unsupervised and Reinforcement Learning.
4. Explain the Machine Learning Pipeline.
5. 🔥 Explain handling of missing values and categorical data encoding.
6. 🔥 Explain feature scaling and train-test splitting.
7. Explain image resizing and grayscale conversion.
8. Explain converting images/video frames into arrays and pixel normalization.

- **Long Answer Questions (6 marks questions)**

1. 🔥🔥 Explain Machine Learning, its types, applications and Machine Learning Pipeline.
2. 🔥 Explain data preprocessing techniques: missing values, categorical encoding, feature scaling and train-test splitting.
3. 🔥 Explain image data preprocessing including resizing, grayscale conversion, array conversion and pixel normalization.

Unit II - Supervised Learning

- **Answers in one or two lines (1 mark questions)**

1. What is Supervised Learning?
2. What is Regression?
3. What is Linear Regression?
4. What is Multiple Linear Regression?
5. What is a Cost Function?
6. What is Gradient Descent?
7. What is Classification?
8. What is Logistic Regression?
9. What is KNN?
10. What is a Decision Tree?
11. What is Naive Bayes?
12. What is SVM?

- **Short Answer Questions (3 marks questions)**

1. 🔥 Explain Simple Linear Regression.
2. 🔥 Explain Multiple Linear Regression.
3. 🔥 Explain Cost Function.
4. 🔥 Explain Gradient Descent.
5. 🔥 Explain Logistic Regression.

6. Explain K-Nearest Neighbours (KNN).
7. 🔥 Explain Decision Tree classification.
8. Explain Naive Bayes Classifier.
9. 🔥 Explain Support Vector Machine (SVM).

● **Long Answer Questions (6 marks questions)**

1. 🔥🔥 Explain Simple and Multiple Linear Regression with suitable examples.
2. 🔥 Explain Cost Function and Gradient Descent in Linear Regression.
3. 🔥 Explain Logistic Regression with its working.
4. 🔥 Explain KNN and Decision Tree algorithms.
5. Explain Naive Bayes Classifier and SVM.

Unit III - Unsupervised Learning, PCA & Image Features

- **Answers in one or two lines (1 mark questions)**

1. What is Unsupervised Learning?
2. What is Clustering?
3. What is K-Means Clustering?
4. What is the Elbow Method?
5. What is dimensionality reduction?
6. What is PCA?
7. Why is PCA used?
8. What is image feature extraction?
9. What is image resizing?
10. What is image data?

- **Short Answer Questions (3 marks questions)**

1. 🔥 Explain Unsupervised Learning and Clustering.
2. 🔥 Explain K-Means Clustering algorithm.
3. 🔥 Explain the Elbow Method for selecting the number of clusters.
4. 🔥 Explain dimensionality reduction and its need.
5. 🔥 Explain Principal Component Analysis (PCA).
6. Explain image data conversion and resizing.
7. 🔥 Explain basic image feature extraction for classification.

- **Long Answer Questions (6 marks questions)**

1. 🔥🔥 Explain K-Means Clustering algorithm with suitable example.
2. 🔥 Explain the Elbow Method and selection of optimal number of clusters.
3. 🔥🔥 Explain PCA and its importance in dimensionality reduction.
4. Explain image feature extraction and preprocessing for classification models.

Unit IV - Neural Networks & Model Evaluation

● Answers in one or two lines (1 mark questions)

1. What is a Neural Network?
2. What is a biological neuron?
3. What is an artificial neuron?
4. What is a Perceptron?
5. What is an activation function?
6. What is the Sigmoid function?
7. What is ReLU?
8. What is a Confusion Matrix?
9. Define Accuracy.
10. Define Precision.
11. Define Recall.
12. What is F1-Score?
13. What is Underfitting?
14. What is Overfitting?
15. What is Cross-Validation?

● Short Answer Questions (3 marks questions)

1. 🔥 Explain Artificial Neuron and Perceptron model.
2. 🔥 Explain activation functions — Sigmoid and ReLU.
3. 🔥 Explain Confusion Matrix.
4. 🔥 Explain Accuracy, Precision, Recall and F1-Score.
5. 🔥 Differentiate between Underfitting and Overfitting.
6. Explain Cross-Validation.

● Long Answer Questions (6 marks questions)

1. 🔥🔥 Explain Neural Networks, Artificial Neurons and Perceptron model.
2. 🔥 Explain Sigmoid and ReLU activation functions with their applications.

3. 🔥🔥 Explain Confusion Matrix and evaluation metrics: Accuracy, Precision, Recall and F1-Score.
4. 🔥 Explain Underfitting, Overfitting and Cross-Validation.
5. Explain different methods used for evaluating Machine Learning models.